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Institutional trading and alternative trading systems[☆]

Jennifer Conrad^a, Kevin M. Johnson^b, Sunil Wahal^{c,*}

^a *Kenan-Flagler Business School, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA*

^b *Aronson+Partners, Philadelphia, PA 19102, USA*

^c *Goizueta Business School, Emory University, 1300 Clifton Road, Atlanta, GA 30322-2710, USA*

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Abstract

We analyze the use of alternative trading systems in a large sample of institutional orders and the trades that constitute these orders. Proprietary data allow us to distinguish between orders and trades filled by day and after-hours crossing systems, electronic communication networks (ECNs), and traditional brokers. Controlling for variation in order and security characteristics, as well as endogeneity in the choice of trading venue, we find that realized execution costs are generally lower on alternative trading systems. Order handling rules and tick size changes implemented in 1997 appear to have reduced the cost advantage of trading on ECNs.

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*Corresponding author. Tel.: +1-404-727-8685.

E-mail address: sunil.wahal@bus.emory.edu (S. Wahal).

1. Introduction

The emergence of alternative electronic trading systems over the past decade has radically altered institutional trading practices. Such trading systems are frequently grouped into two categories: crossing systems (such as ITG's POSIT) in which institutional orders are brought together and "crossed" at some prevailing price, and electronic communication networks (ECNs), such as Instinet, that allow counterparties to trade anonymously at negotiated prices. These systems have experienced rapid growth and appear to provide a significant source of competition for order flow, for both exchanges and dealer markets. For instance, the nine currently operating ECNs account for almost 40% of the dollar volume of trading in Nasdaq securities (SEC, 2000).¹ Similarly, trading volume on POSIT, the most prominent crossing system, grew at a compounded annual growth rate of over 45% between 1988 and 1999, and recorded a total trading volume of 6.5 billion shares in 1999.

The management of trading, and by extension the choice of trading venue, is an important issue for institutions. Institutional investors must, by law, seek "best execution" of their trades, and execution costs are an important determinant of investment performance. However, as Macey and O'Hara (1997) note, best execution is a multidimensional concept encompassing market impact, timing, trade mechanism, anonymity, commissions, and even quid pro quo arrangements. Alternative trading mechanisms offer different combinations of these aspects of execution quality and allow investors to pursue tradeoffs between such attributes.

In addition to its obvious importance for practitioners, and for research in market microstructure, an understanding of alternative trading systems is also relevant from a regulatory perspective. A proliferation of such systems can cause fragmentation of markets, with some investors trading at prices unavailable (or invisible) to others. Recognizing this possibility, in 1997 the SEC introduced order handling rules requiring market makers to reflect ECN quotations in their prices, and continues to be concerned with fragmentation (SEC, 2001). Continued growth in such systems also prompted the adoption of Regulation ATS in 1998, requiring trading systems to choose between being a market participant (e.g., broker) and registering as an exchange.

Given the growth in trading volume on alternative trading systems and the regulatory issues involved, academic interest in their use has increased. However, because such systems are proprietary, their trades are unobservable in conventional data sources such as TAQ. In this paper, we use unique proprietary data on the release of \$1.6 trillion in equity trades by institutional investors, to provide a glimpse of the execution services offered by such systems. The sample consists of 797,068 orders submitted by 59 institutions between the first quarter of 1996 and the first quarter of 1998. The data show each individual order, the trades resulting from the order, and most important, the use of alternative trading systems in executing these

¹ The nine ECNs are: Attain, Brut, Instinet, Island, Bloomberg Tradebook, GFI Securities, Market XT, NexTrade, and Arca/Redibook.

trades. Each trade is uniquely identified with the broker or alternative trading system responsible for executing the trade.

We use these data to separate orders into those filled entirely by one trading system (single mechanism orders) versus those in which trades are executed by more than one trading system (multiple mechanism orders). We then separate single mechanism orders into one of four types, which one can think of as representing a discrete ordering by how aggressively the order is marketed. The first two types consist of orders that cross via a passive call market. These crosses can take place during the trading day on POSIT or after-hours on Instinet's crossing system. Such crossing systems anonymously match buys and sells in a liquidity pool confined to participants within the trading system. Therefore, they attract orders with lower immediacy requirements, and provide no price discovery mechanism. The third category is orders executed on ECNs. In our data, this category consists entirely of orders executed on Instinet's "day-system" which allows traders to anonymously enter orders on-screen and trade with other institutions, broker-dealers, market makers, and exchange specialists. (Other ECNs catering to institutional trading, such as Tradebook, did not commence operations until the fourth quarter of 1998.) ECNs provide more immediacy than crossing systems in that they permit counterparties to anonymously negotiate, and therefore provide a price discovery mechanism. Finally, orders executed through the traditional brokerage system on the NYSE, Amex, or Nasdaq are referred to as broker-filled orders. Broker-filled orders are the most aggressive in providing immediacy since brokers actively search for counterparties across all trading venues and may even commit capital. This last category serves as the numeraire for our tests.

We find that crosses take place largely in listed securities whereas ECNs focus on Nasdaq stocks. This separation is natural, given the trading mechanisms employed by these systems. Since crossing systems provide no price discovery mechanism, they need at least two attributes to provide attractive fill rates. First, they need good primary-market price discovery to support the crossing system. The NYSE provides such a mechanism; see [Hasbrouck \(1995\)](#) for evidence. Second, they need a minimum threshold of trading volume by participants on the system so that the pool of liquidity is sufficiently large. ECNs, on the other hand, provide active price discovery—a feature that is particularly important when the primary market is relatively fragmented and has high bid-ask spreads. Therefore, Nasdaq stocks provide a natural setting in which ECNs can compete with market makers for order flow. Our data also suggest important differences in the characteristics of orders across trading systems. For example, the average dollar volume of orders executed by crossing systems and ECNs is substantially lower than for those executed by brokers. In general, broker-filled orders are larger, broken up into more trades, and have higher fill rates than orders sent to alternative trading systems.

To examine the competitive performance of alternative trading systems, we use the implementation shortfall method of [Perold \(1988\)](#) and [Keim and Madhavan \(1995, 1997\)](#). This method characterizes total execution costs as the sum of implicit (price-movement-related) and explicit (commission) execution costs. We use two methods of controlling for differences in order and security characteristics. First, we use a

matched-sample approach that controls for trade direction (buy versus sell), order instruction, order size, exchange listing, and market capitalization, but does not impose linearity or functional form restrictions. We find that crosses have total execution costs that are substantially lower than those provided by traditional brokers. On average, the cost difference between day crosses and broker-filled orders is approximately 30 basis points. For the average-priced stock crossed during the trading day in our sample, this corresponds to approximately 13 cents per share. The total execution cost difference between broker-filled and ECN-executed orders is higher, at 66 basis points, which corresponds to an average of 20 cents per share for stocks traded on ECNs. Most of the difference in execution costs is attributable to the prices at which trades are executed, rather than commissions. The second method of controlling for differences in order characteristics is an ordinary least squares regression analysis. The regression approach allows us to include institution-specific fixed effects, which is important because [Keim and Madhavan \(1997\)](#) show that institutions have different approaches to filling orders that are reflected in their execution costs. Consequently, estimates of cost differentials are more conservative. Using such regressions, we estimate the execution cost differential between day crosses and broker-filled orders to be 17 basis points for buys and 14 basis points for sells. For ECN-executed orders, the differentials are 28 basis points for buys and 54 basis points for sells.

It is possible that the choice of trading mechanism is endogenous to (ex post) realized execution costs, particularly since each trading mechanism reflects a different level of effort on the part of the trader. We address the endogeneity issue by using an endogenous switching regression method described by [Madhavan and Cheng \(1997\)](#) that controls for selectivity. The procedure involves estimating a first-stage probit regression predicting the choice of trading mechanism, which then feeds into a second-stage execution cost regression. Using this method, however, does not change our basic conclusion; in general, the cost differentials described above persist, and in some cases widen. As in the OLS regressions, the differentials narrow when institution-specific fixed effects are introduced. We also estimate the switching regressions using a matched sample of orders that ensures a common level of support across orders sent to competing trading platforms (see [Heckman et al., 1997](#)) and obtain similar results.

Finally, we focus our attention on orders for which selectivity is likely to be largest, that is, orders that employ different trading mechanisms for constituent trades. Interestingly, we find that although brokers are the last method of execution in a majority of the orders, a substantial fraction (over 40%) of orders have their last trade executed on an alternative trading system. If the last portion of the trade is the most difficult, this suggests that the dealer market is not always the “market of last resort.” We conduct a trade-level analysis of such orders and again find that, controlling for both order and trade-level characteristics, execution costs are lower for crossing systems and ECN-executed trades.

Given the assumption in the literature that investors seek to minimize execution costs, as well as the fiduciary burdens placed on institutional investors in particular, differences in average “excess” costs across trading platforms should be zero. Thus,

the intriguing question is how to interpret the evidence in our sample. It is possible that the cost differentials are not real, but simply due to our inability to perfectly account for order characteristics, market conditions, and/or selectivity. While no system of controls can be perfect, our results are robust across a variety of methodologies and econometric techniques. It is also possible that differences in execution costs are a temporary “disequilibrium.” Indeed, the large differences in order characteristics across trading platforms, combined with the remarkable growth in the number of and volume on alternative trading systems, is consistent with a movement toward equilibrium. If lower execution costs are responsible for the dramatic growth in volume on these systems, then we would expect competition for order flow, changes in industry structure, and perhaps changes in regulatory policy to reduce these differences over time. Indeed, we see two instances of such changes in our sample, both related to ECN trading. First, order handling rules introduced in 1997 required broader dissemination of prices posted on ECNs, thereby reducing fragmentation between Nasdaq and ECNs. Second, the benefit of the availability of finer price increments on ECNs but not on exchanges was also reduced by the SEC-mandated reduction in quoted tick sizes in 1997. Subsequent to both of these events, we find a decline in the relative use of ECNs by the institutions in our sample. There also appears to be a decline in total ECN trading volume after the tick size change. More telling is the evidence that the cost advantage of trading on ECNs also declines after each of these events, although it remains nonzero. This suggests that if the cost differentials for ECNs in our sample represent a “disequilibrium,” then continued competition and changes in regulatory policy might further reduce them.²

Changes in the order handling rules and tick size do not appear to have affected trading on crossing systems. Although the estimated differentials associated with these systems are smaller, there is another explanation for their lower costs that does not require a disequilibrium argument. Specifically, the opportunity cost of *not* filling an order can be particularly large for crossing systems because they are passive in nature. Moreover, our estimates of such opportunity costs in our sample is undoubtedly biased downwards because single-mechanism orders that are sent to crossing systems but are completely unfilled do not appear in our data. Thus, our evidence in these cases is based on realized execution costs and is therefore conditional on the execution of trades. Although opportunity costs are notoriously difficult to measure, Edwards and Wagner (1993) estimate that they may range from 16 basis points for large-cap stocks to 220 basis points for small cap stocks, a magnitude large enough to account for the cost differential between crossed and broker-filled orders.

²Changes in industry structure, as opposed to regulatory policy, are also consistent with this argument. The increase in competition is evident in the increase in the number of ECNs over our sample period. It can also be observed in the response of exchanges and dealer markets to incursions by alternative trading systems. For instance, the NYSE recently introduced Institutional XPress, designed specifically to attract institutional order flow. Similarly, Nasdaq’s SuperMontage seeks to compete for institutional orders by encapsulating features akin to those provided by ECNs, such as anonymous postings. See “Plan to upgrade Nasdaq trading passes the SEC,” *Wall Street Journal*, 1/11/2001.

Our paper is part of a small but growing literature that examines the impact of technology-based innovations in the trading environment. [Hendershott and Mendelson \(2000\)](#) develop a theoretical model to explore the consequences of the introduction of a crossing network on the underlying market. Other empirical papers examine the role of ECNs in price discovery. [Huang \(2002\)](#) uses quote data from the Nastroq database and concludes that ECNs are important contributors to price discovery. [Barclay et al. \(2002\)](#) examine ECN trades and find that average quoted, realized and effective spreads are smaller for ECN trades than for market makers; see also [Barclay and Hendershott \(2000\)](#) for an analysis of after-hours trading, some of which takes place on ECNs. In these studies, however, the analysis focuses on all trades occurring on a trading platform; the authors cannot distinguish between single mechanism and multiple mechanism trades, nor can they identify the sequence of the trade in the underlying order. In contrast, [Fong et al. \(2001\)](#), like us, are able to identify the order from which trades are generated. They analyze data from the Australian Stock Exchange, which is itself electronic, and find that the magnitude of off-market trading (defined as the total of trading volume on ECNs, crossing systems, and upstairs markets in Australia) is driven by institutional trading interest and liquidity. However, Fong et al. do not analyze multiple mechanism orders and do not investigate sequencing in the use of trading platforms to fill such orders.

To our knowledge, this study represents the first comprehensive analysis of data on institutional investors' strategic use of multiple trading platforms in minimizing execution costs. There are, however, two related studies. [Domowitz and Steil \(1999\)](#) examine trading data from a single anonymous institution and compare execution costs across trading systems. They find that electronic systems provide lower execution costs than market makers on Nasdaq. They also find that while market impact costs are comparable across electronic systems and exchanges for NYSE stocks, total execution costs (including commissions) of trades executed on the electronic systems are lower. [Naes and Odegaard \(2001\)](#) also examine the trades of a single institution, the Norwegian Petroleum Fund. In a study of 4,200 orders that are sent first to crossing systems and then to brokers, they find that execution costs of crossed trades are lower. They also present evidence that the cost of failing to execute on the crossing system can be significant.

Our paper proceeds as follows. In Section 2, we provide some brief institutional details. In Section 3, we describe our data. We present results in Section 4 and conclude in Section 5.

2. Institutional details

In this section, we present a brief description of the rules and features of crossing systems and ECNs used by institutional investors. In crossing systems, traders enter unpriced (buy or sell) trades, which are crossed at pre-specified times, at prices determined in the security's primary market. Since the trades are unpriced, such systems do not provide a direct price discovery mechanism. Moreover, immediacy is

not guaranteed since crosses take place only at a few discrete points in time, and if there is an order imbalance, the trade may not be filled. Crossing services to institutions are provided by three entities. First, the New York Stock Exchange's Crossing Session I crosses trades in individual stocks between 4:15 p.m. and 5:00 p.m. based on the NYSE closing price. Second, Instinet Crossing (not to be confused with Instinet's "day" ECN system) crosses listed stocks at the closing price on the NYSE, and Nasdaq stocks at the closing inside quote midpoint. Instinet Crossing is an after-hours cross with no preset time, but trades usually take place between 7:00 p.m. and 8:00 p.m. Finally, the most prominent crossing system is ITG's POSIT. Currently, POSIT crosses trades eight times during the trading day—at 9:40 a.m., 10:00 a.m., 10:30 a.m., and hourly from 11:00 a.m. to 3:00 p.m.

Each cross takes place at a randomly selected time within a five-minute window immediately following the scheduled time. As mentioned earlier, we distinguish between day crosses (on POSIT) and after-hours crosses (on Instinet) because there are substantial differences in market conditions, crossing prices and the availability of other trading venues. In day crosses, the prevailing quote midpoint in the stock's primary market serves as the crossing price, whereas in after-hours crosses, the closing price of the security serves as the crossing price. In an analysis of after-hours trading, [Barclay and Hendershott \(2000\)](#) find that the majority of after-hours trading in NYSE stocks takes place at or near the closing price. Commissions on all crossing systems are usually fairly small, from one to two cents a share.

The number of crosses on POSIT has increased rapidly since its inception in 1987. Over our sample period, the number of crosses increased from four (in early 1996) to five (in late 1996). A sixth match was added in the fall of 1998, a seventh match in May of 2000, and the eighth match was added in March of 2001. Consistent with our interpretation that crossing systems rely on a pool of liquidity in the underlying market, the timing of the new matches follows closely the periods in which the NYSE experiences heavy volume. [Mendelson \(1982\)](#) shows how liquidity and trading volume interact in a batch "clearing house" system similar in spirit to POSIT.

Unlike crossing systems, ECNs allow traders to enter priced trades in a variety of forms in a screen-based system. The buyer and seller remain anonymous during the trading process and trade execution reports list the ECN as the contra-side party, thereby maintaining post-trade anonymity. Since trading on ECNs uses priced orders, such systems have the potential to contribute to price discovery over and above the stock's primary market. In fact, [Huang \(2002\)](#) finds that ECNs are important contributors to price discovery and are the dominant trading venue in eight of the ten most active stocks that he analyzes. Such systems afford traders a variety of features that provide considerable flexibility in managing execution costs. For instance, negotiation features permit users to conduct anonymous negotiations over price and size. Similarly, reserve size orders can be displayed such that only part of the available size is shown, facilitating the breakup of orders. As mentioned earlier, of the nine currently operating ECNs, Instinet is the only active one in our sample. The remaining ECNs commenced operations after the end of our sample period. Commissions on Instinet vary, and are usually determined by the amount of trading volume the institution contributes.

The traditional method of executing institutional trades is through the well-known broker system. For NYSE stocks, orders can be sent directly to the NYSE floor, where trades can be executed with or without specialist intermediation. Large orders can also be executed in the upstairs market, where brokers search for liquidity and prices are determined by negotiation (see Hasbrouck et al., 1993; Keim and Madhavan, 1996; Madhavan and Cheng, 1997). For Nasdaq stocks, large institutional orders are typically handled by full service or dedicated brokers. These brokers find counterparties and negotiate a price to fill a trade, much like the NYSE upstairs market. Trading using traditional brokers is not anonymous and can result in leakage of information. Commissions are negotiated and depend on the volume generated by the institution.

3. Data sources and procedures

Our primary data are provided by the Plexus Group, a consulting firm to institutional investors that monitors the costs of institutional trading. The data include each order and the trades (transactions) resulting from that order. Order-level information is *ex ante* and includes the following: a ticker symbol, the market capitalization of the stock, a buy–sell indicator that is based on the institution initiating the trade, the date when the trading decision was made, the closing price on the day before the decision date, the geometric mean of daily trading volume over the five days prior to the decision date, and an instruction variable indicating if the order is a market order, a limit order, or a cross. Each order can result in one or more trades. For each trade, the data show the number of shares bought or sold, the price at which the trade is executed, a broker ID indicating the broker responsible for executing the trade, and the commissions charged for the transaction (in cents per share). Institution names are removed from the database and replaced with institution codes to preserve confidentiality. Institution codes are mapped by Plexus into one of three investment styles: momentum (which includes any short-term technical trading), value, and diversified (which includes quantitative styles, including indexing, that are neither technical nor value in nature). The data are similar in structure to those used by Keim and Madhavan (1995, 1997) and Conrad et al. (2001) but represent a more recent and longer time series. Our sample comprises orders submitted by subscribing institutions from the first quarter of 1996 to the first quarter of 1998. For a subset of tests, we also use volume data provided by Instinet Inc. Specifically, for a sample period matching with our Plexus data, Instinet provided us with the total number of shares traded on the Instinet “day” system for each security, on each calendar day.

We apply several filters to the raw data provided by the Plexus Group. We follow Keim and Madhavan and eliminate orders for stocks trading under \$1.00 and discard all non-U.S. stocks. Orders for which the broker instruction (i.e., market order, limit order, or cross) field is either missing or equal to ‘X’ are also removed from the sample. This last filter has the effect of removing eight institutions from our sample, mostly from the latter part of the time series. At the request of the Plexus

Group, we also remove an institution for which the data are suspect. Although investment style information is missing for 14 institutions, we do not remove these from the sample since the remaining information on orders and trades is complete. Over time, new institutions enter the sample as they are added to Plexus's client base, and some existing institutions leave the database. We do not require a continuous presence for an institution to be in the sample. These data filters result in a final sample of 797,068 orders, submitted by 59 unique institutions which generate approximately 2.15 million trades. Order-level data are matched with price and return information from the Center for Research in Security Prices.

Based on data-reporting protocols and guidance from the Plexus Group, we apply a classification algorithm that categorizes all trades into one of four categories: day crosses, after-hours crosses, ECN-executed trades, and broker-filled trades. Details of this classification procedure are reported in the appendix.³

While each *trade* can only fall into one of the four categories, an *order* can be filled by trades from multiple categories. Issues of execution quality are more appropriately addressed at the order, rather than the trade-level, so it is important to distinguish between orders executed using one or multiple mechanisms. If an order is filled entirely by trades executed in just one of the four categories, we refer to it as a “single mechanism” order; if more than one category is employed, we refer to it as a “multiple mechanism” order.

Table 1 shows the distribution of single and multiple mechanism orders. Of the total sample of 797,068 orders, approximately 90% (723,998) are single mechanism orders. These are unevenly distributed across the four categories; the majority (560,712 orders) are filled by traditional brokers, 108,111 orders by day crosses, 4,048 by after-hours crosses, and 51,127 by ECNs. Although we continue to report results for after-hours crosses throughout the paper, given the small sample size, the statistical precision of the estimates is unlikely to be high. There appears to be considerable time-series variation, but no trend, in the distribution of single mechanism orders. For day crosses, the number of orders per quarter generally ranges between 10,000 and 20,000. However, there are two notable exceptions. The third quarter of 1996 and the second quarter of 1997 contain only 2,183 and 1,650 such orders. This is because data for one institution that is a frequent user of crossing services are not available for these two quarters. The number of ECN-executed orders generally ranges between 4,000 and 6,000 per quarter. The apparent slight decline in the number of ECN executed orders over time is primarily because institutions with different propensities of ECN use enter and exit our sample over time. If we restrict our sample to institutions with complete data coverage over the entire time series, the frequency of ECN use for our sample institutions does not change. We examine total ECN volume, as well as ECN trading by the institutions in our sample, in Section 4.7.

³ An earlier version of this paper also reported information another category, internal crosses. Unfortunately, the accuracy of the data for this category are suspect and the results are no longer reported.

Table 1

Distribution of single and multiple mechanism orders

Single mechanism orders are those in which all the trades in an order are executed in only one of the four categories listed below. Multiple mechanism orders are those in which more than one of the four trading mechanisms is used to fill the order. Day crosses refer to orders executed on ITG POSIT and after-hours crosses refer to orders executed on Instinet's after-hours Crossing Network. Orders executed on Instinet's anonymous trading system are labeled ECN. Broker-filled orders are those that are filled via traditional brokerage services.

Quarter	Single mechanism orders				Multiple mechanism orders
	Day cross	After-hours cross	ECN	Broker filled	
961	11,466	523	5,596	41,082	7,364
962	6,816	617	6,400	66,642	9,844
963	2,183	634	6,700	42,630	9,116
964	10,351	631	6,618	70,954	10,316
971	17,441	496	6,379	76,828	8,107
972	1,650	298	6,487	79,332	8,955
973	23,496	428	5,115	58,736	6,311
974	18,693	140	4,967	77,060	6,872
981	16,015	281	3,315	47,448	6,185
All	108,111	4,048	51,127	560,712	73,070

4. Results

Panel A of Table 2 reports statistics on various measures of order size, order difficulty, order type, and security characteristics. We use two measures of order size: dollar order size (measured in thousands of dollars) and relative volume (defined as the number of shares traded, divided by the geometric mean of daily trading volume over the previous five days). Using both measures, the data show substantial differences in order size between the categories. The average size of orders sent to crossing systems and ECNs is considerably lower than those worked by traditional brokers. The average order executed via a day cross is worth \$179,000. The corresponding average for the ECN is \$194,000 and for broker-filled orders, is \$1.4 million. For multiple mechanism orders, the average order size is also high (\$2.0 million). While means are obviously skewed by particularly large orders, the same pattern is also evident in medians. Measured relative to average trading volume, broker-filled orders are still significantly larger than crosses, but are comparable to ECN-executed orders.

In most models of information-based trading, such as Kyle (1985), breaking up an order allows the trader to maximize his/her gains from information. From the perspective of minimizing execution costs, larger and/or more difficult orders are likely to be broken up so as to minimize the price impact of constituent trades (Bertsimas and Lo, 1998). The evidence in our sample is consistent with this. The average number of trades in the day crosses, after-hours crosses and ECN categories are 1.29, 1.43, and 1.53 respectively; for broker-filled orders, this rises to 2.19 trades

Table 2

Order and security characteristics

Panel A shows characteristics of various types of orders. Single mechanism orders are those in which all the trades in an order are executed in only one of the four categories listed below. Multiple mechanism orders are those in which more than one of the four trading mechanisms is used to fill the order. Average trading volume is the geometric mean of daily trading volume five days prior to the decision date. Relative volume is defined as the number of shares in the order divided by average volume. The fill rates are computed by calculating the percentage of the order that is filled. Panel B shows probit regressions of order type on various explanatory variables. The dependent variable is equal to one for market orders and zero for limit orders. In addition to the coefficient estimates and the associated *p*-values (in parentheses), the panel also shows marginal effects (implied changes in probabilities) in square brackets. For indicator variables, the marginal effects are computed assuming a change from zero to one. For continuous variables, the marginal effects are computed assuming a one standard deviation change from the mean.

Panel A: Descriptive Statistics

	Single mechanism orders				Multiple mechanism orders
	Day cross	After-hours cross	ECN	Broker filled	
Order size (\$000)					
10th percentile	6	13	9	8	70
Mean	179	380	194	1,474	2,005
Median	44	110	53	137	587
90th percentile	387	894	360	2,821	4,768
Relative volume	0.06	0.17	0.35	0.30	0.67
Number of trades	1.29	1.43	1.53	2.19	4.07
Fill rates					
% Partial fill	7.91	6.87	5.90	3.94	16.24
% Complete fills	92.09	93.13	94.10	96.06	83.76
# Buy orders	69,124	2144	34,371	286,908	40,922
# Sells orders	38,987	1904	16,756	273,804	32,148
\$ Buys/\$ Sells	1.20	1.01	1.34	0.98	
NYSE/Amex	98,258	3027	12,971	425,002	44,474
Nasdaq	9,853	1021	38,156	135,710	28,596
Trading volume	7,679	5,100	3,022	7,120	6,022
Market value (\$M)	12,909	9,527	3,803	11,122	9,239
Market orders (%)	—	—	64	97	—
Limit orders (%)	—	—	36	3	—

Panel B: Probit regressions of order type

	ECN orders	Broker-filled orders
Intercept	1.1923 (0.000)	0.4060
Log (relative volume)	0.0014 [0.00] (0.707)	−0.0468 [−0.00] (0.000)

Table 2 (*continued*)

	ECN orders	Broker-filled orders
Buy/sell indicator	−0.7339 [−0.21] (0.000)	−0.3671 [−0.01] (0.000)
Log (market cap)	0.1581 [0.05] (0.000)	0.1689 [0.00] (0.000)
Diversified style indicator	−2.8689 [−0.63] (0.000)	−0.5796 [−0.01] (0.000)
Value style indicator	−2.5416 [−0.79] (0.000)	0.7429 [0.00] (0.000)
Exchange indicator	0.5871 [0.16] (0.000)	−0.5541 [−0.01] (0.000)
Cumulative return	0.1537 [0.05] (0.000)	0.0834 [0.00] (0.000)
Pseudo- R^2	0.26	0.14

per order. For multiple mechanism orders, 4.07 trades are required to fill the order on average. In fact, although multiple mechanism orders represent only 10% of the total number of orders, they represent 47.5% of dollar trading volume. Since they are an important component of our sample in both type and volume, we examine them separately in Section 4.6.

Fill rates also vary. For crosses, the fill rate is a function of liquidity on the system or prevailing counterparty depth, a factor that is exogenous to the trader. In contrast, traders using an ECN can negotiate anonymously and therefore endogenously increase the probability of a complete fill. Similarly, for orders filled by traditional brokers, prices can be negotiated, thereby increasing fill rates. Consistent with this, the percentage of complete fills is approximately 92% for day crosses but rises to 94% and 96% for ECN and broker-filled orders, respectively. Multiple mechanism orders represent the most difficult of all categories, and have a complete fill rate of only 81%. We note, however, that fill rates, particularly for crossing systems, should be interpreted with caution. The 92% fill rate for day crosses is based on single-mechanism orders only. Inclusion of multiple mechanism orders will clearly lower this fill rate. Moreover, as mentioned earlier, completely unfilled single mechanism orders sent to crossing systems would not appear in our sample.

The data also show clientele effects in which crossing systems and ECNs compete for order flow in different dimensions. Extant studies document that transaction costs are lower on the NYSE than on Nasdaq (see Bessembinder, 1997, 1999; Huang and Stoll, 1996; Chan and Lakonishok, 1997) and that the NYSE provides the vast majority of price discovery for listed securities (see Hasbrouck, 1995). Since crossing systems do not provide price discovery, they compete for order flow by allowing traders to (passively) trade with each other. In other words, crossing systems require that the primary market provide adequate price discovery and that the system attract a minimum threshold of volume from the primary market (Hendershott and

Mendelson, 2000). The higher the primary market volume, the more likely it is that the crossing system will attract order flow. Our results are consistent with these arguments. In our sample, over 90% of all orders executed using day crosses are for NYSE securities. These securities have an average market capitalization of \$12.9 billion and an average trading volume over a five-day period prior to the submission of the order of 7.6 million shares. The statistics for after-hours crosses are similar. Almost 75% of orders executed using after-hours crosses are for NYSE stocks and the average market capitalization in these stocks is \$9.5 billion.

In contrast, ECNs allow for active price discovery and can therefore compete with primary markets in which transaction costs are higher and order flow is fragmented. High primary-market trading volume is not a prerequisite for ECNs since traders can negotiate prices. Consistent with this, almost 80% of all ECN-executed orders in our sample are for Nasdaq securities. These stocks have a much lower market capitalization (\$3.8 billion) and average trading volume (3.0 million shares) than for crossing systems.

There appear to be some differences in the number and dollar value of buys and sells across the categories. The number of buy orders is approximately twice as large as the number of sell orders. Measured in dollars, the difference is smaller: the dollar buy–sell ratio is 1.20 for day crosses, 1.34 for ECNs, and 0.98 for broker-filled orders. Across a trading system as a whole, the dollar value of executed buys has to equal the dollar value of executed sells.

Therefore, any divergence of this ratio from one for our sample reflects the buy–sell propensities of the institutions specific to the Plexus database. For example, the institutions that use crossing systems and ECNs in our sample could simply be gaining assets over this period and therefore buy more often than sell.⁴ Other institutions or intermediaries could have different buy–sell propensities. Regardless, because other authors (e.g., Keim and Madhavan, 1997; Chan and Lakonishok, 1995) find systematic differences in the execution costs of buy and sell orders, we separate buys and sells in our matching analysis and regressions.

Finally, we report information on the types of orders used by institutions in our sample. Keim and Madhavan (1995) report four types of orders used by institutions. From most active to most passive, these are market orders (including “not-held” market orders), working orders, crossing orders, and limit orders. In not-held orders, the broker has discretion with respect to the pricing and timing of trades. Working orders are similar in that brokers are given such orders to execute over time while minimizing price impact. Our data contain three of these order types: market orders

⁴There are other institutional considerations that could also influence buy–sell propensities. For example, when an institutional money manager is hired, the new manager is more sensitive to execution prices (since they influence his/her investment performance) and might use alternative trading systems. In contrast, when a money manager is terminated, brokers may be used to quickly liquidate or cash out portfolio positions; such managers may be less sensitive to trading costs. This could explain the greater fragmentation observed in buy orders in our sample. Another consideration is that since execution uncertainty is large for crossing systems, institutions are less willing to use such systems for sell orders. If the buy orders are executed but the sells are not, the institution could end up being more than 100% invested. Greater fragmentation in buys could also occur because buys are more expensive than sells.

(including not-held orders), crossing orders (identified as those crossed on either a day or after-hours cross), and limit orders. Since crosses are identified separately, we report the distribution of market versus limit orders for ECN-executed and broker-filled orders. The use of limit orders is much larger on ECNs (36%) than for broker-filled orders (3%).

4.1. The determinants of order type

In addition to the type of trading mechanism selected by an investor, the choice of order type also reflects the aggressiveness of trading. Keim and Madhavan (1995) examine the use of limit orders by institutional investors. They find that buy orders are more likely to be submitted as passive limit orders, that limit orders are more likely in smaller, less liquid stocks and exchange-listed stocks, and that “value” managers are more likely to use limit orders. Given the substantially higher frequency of limit orders in our sample of ECN trading, we estimate probit models that predict the choice of order type separately for ECN and broker-filled orders.

We set the dependent variable equal to one for a market order and zero for limit orders. The explanatory variables in the models are similar to those in Keim and Madhavan (1995) and include order- and security-specific information, as well as the investment style of the institution. The order-specific information (such as the logarithm of relative volume and a buy–sell indicator) captures the idea that order characteristics can be reflected in the choice of order type. Security-specific information (such as the exchange indicator and the cumulative return) is included because some types of securities might be harder to trade via limit orders. Investment style indicators reflect the aggressiveness of the institution (perhaps because of the nature of information inherent in the trade), which could translate into the choice of an aggressive or passive order.

The results of these probit models are presented in Panel B of Table 2. In addition to coefficient estimates, we also report the marginal effects (changes in probabilities) in square brackets. For indicator variables (exchange listing, buy–sell, and investment style indicators), the change in implied probabilities is computed assuming a change in the independent variable from zero to one. For the continuous variable (the volatility of return), the change in probability is computed assuming a one standard deviation change from the mean of the variable.

In general, our results are consistent with those of Keim and Madhavan (1995). For the ECN sample, the model has good explanatory power (with a pseudo- R^2 of 0.26) and the probability changes associated with these variables are generally large. Buy orders are more likely to be executed through limit orders, while market capitalization and cumulative return are positively related to the use of market orders. Also, passive investment styles (such as “value”) are less likely to use market orders. We find that ECN trading in exchange-listed stocks is more likely to be done through market orders. This contrasts with Keim and Madhavan’s evidence that exchange-listed stocks are more frequently traded using passive strategies. This difference should be treated cautiously, however, as the *total* level of ECN trading in exchange-listed securities is quite small in our sample. Overall, despite the difference

in the overall propensity to use limit orders in our sample of ECN trading, the factors that determine their use appear to be similar across ECNs and exchanges.

In the broker-filled order sample, only 3% of the orders represent limit orders. As a result, the regression may suffer from a lack of power and indeed, the pseudo- R^2 is lower (0.14). Even though some variables are similar in sign and statistically significant, the associated changes in probabilities are extremely small—in all cases, less than 1%.

4.2. Measurement of execution costs

Given trade, order, and decision-level data, we follow well-established implementation shortfall methods for measuring execution costs (Perold, 1988; Keim and Madhavan, 1997). Order-level implicit and explicit trading costs are calculated as follows:

$$\text{Implicit Cost} = \frac{P_t}{P_d} - 1, \quad (1)$$

$$\text{Explicit Cost} = \frac{C_t}{P_d}, \quad (2)$$

where P_t is the trade-volume-weighted average price at which the order is executed, P_d is the closing price on the day before the decision is made to buy or sell, and C_t is volume-weighted commissions per share. As Keim and Madhavan (1997) point out, since P_d represents an unperturbed price prior to the submission of the order, the implicit execution cost represents the difference in performance between the returns to a paper portfolio versus a portfolio constructed from realized transaction prices. Total execution costs are simply the sum of the implicit and explicit costs defined above.

Given the distribution of fill rates described earlier, it is important to adjust partially filled orders for opportunity costs (see Edwards and Wagner, 1993). For the partially filled portion of an order, we compute opportunity costs as in Eq. (1), but replace P_t with a closing price ten days after the decision date (P_{+10}). This is tantamount to “closing the books” on the order at this time; the choice of the ten-day window corresponds to the 95th percentile in the distribution of the number of days required to fill an order. The implicit cost is weighted by the fill rate, and the opportunity cost is weighted by one minus the fill rate. As a robustness check, we also close the books on orders 15 days after the decision date (corresponding to the 99th percentile in the distribution of the number of days required to fill an order). The results are similar and not reported in tables.

Costs calculated as above are at the *order* level. Thus they are different from trade- or transaction-level costs computed from conventional market microstructure databases but similar to the trade-package costs reported by Chan and Lakonishok (1995, 1997) and the order-level costs presented in Keim and Madhavan (1995, 1997) and Conrad et al. (2001). While these order-level costs are appropriate descriptors for single mechanism orders, they present a problem for multiple mechanism orders

because we cannot attribute the cost to a particular trading mechanism. As a result, the majority of our analysis focuses on single mechanism orders. We return to an analysis of multiple mechanism orders in Section 4.6.

The data show large differences in execution costs across the categories. Multiple mechanism orders, evidently the most difficult to fill based on evidence in Table 2, have the largest implicit costs (71 basis points), followed by broker-filled orders (43 basis points). The average implicit cost of ECN-executed orders is 19 basis points and the implicit cost of day crosses is substantially smaller at 12 basis points. Of course, this rank ordering of implicit costs is inversely related to the rank ordering of order difficulty. Therefore, we proceed to execution cost comparisons that control for security characteristics and order difficulty.

4.3. Matched sample execution cost comparisons

As mentioned earlier, the choice of trading venue represents varying degrees of aggressiveness on the part of the institution. Day and after-hours crosses are the most passive since orders are exposed only to a fixed liquidity pool. ECN executions are more aggressive in that they allow for negotiations on price and quantity. The most aggressive is the use of traditional brokers, who are sometimes willing to commit capital in the search for best execution. These differences result in a natural sorting of order difficulty across the categories, and are evident in the results in Table 2. Therefore, any comparison of execution costs must, at a minimum, account for variation in order difficulty. In addition, there can also be substantial differences in the characteristics of the securities traded, which can influence liquidity and hence trading costs. In this section, we employ a matched-sample methodology to control for order and security characteristics.

We match each crossed and ECN-executed order with broker-filled orders on the following dimensions: exchange listing (NYSE/Amex versus Nasdaq), buy or sell, and order instruction (market or limit) for ECN-executed orders. We also match on relative volume and market capitalization. The first three matching criteria require exact matches but for the last two, we allow a 10% tolerance (i.e., the market capitalization and relative volume of broker-filled orders have to be between 90% and 110% of the ECN or crossed order). This procedure allows us to match 99% of crossed orders and 90% of ECN-executed orders. We compute implicit, explicit, and total execution cost differentials as the execution cost difference between the primary order and the matched broker-filled orders. If more than one broker-filled order fulfills the matching criteria, the average execution cost of the broker-filled (matched) orders is subtracted.

Table 3 shows average matched-sample execution cost differentials in percent, with standard errors in parentheses. The cost differentials for day crosses are large, statistically significant, and robust across sample periods. The average implicit cost difference for the entire sample is –20 basis points, while the average explicit cost difference is –10 basis points, resulting in a total cost difference of –30 basis points. Although there is quarterly variation in total cost differentials, most of this variation comes from implicit costs; explicit cost differences remain relatively flat at 9–11 basis

Table 3

Matched sample execution cost differentials

Each order executed via a day cross, after-hours cross, or ECN is matched with orders executed by brokers on the following dimensions: exchange listing (NYSE/Amex vs. Nasdaq), buy or sell, order instruction (market or limit order), relative volume (with a 10% tolerance), and market capitalization (with a 10% tolerance). Where more than one broker-executed order matches, the average execution cost of all matched (brokerage) orders is calculated. The table shows the average implicit, explicit, and total execution cost differential between the matched sample and the orders of interest. All costs are reported in percent. Standard errors appear in parentheses.

	Day cross			After-hours cross			ECN		
	Implicit	Explicit	Total	Implicit	Explicit	Total	Implicit	Explicit	Total
961	−0.29 (0.02)	−0.09 (0.00)	−0.37 (0.02)	−0.19 (0.13)	−0.11 (0.00)	−0.30 (0.13)	−0.94 (0.08)	0.06 (0.00)	−0.86 (0.08)
962	−0.19 (0.02)	−0.09 (0.00)	−0.28 (0.02)	−0.23 (0.10)	−0.11 (0.00)	−0.34 (0.10)	−0.76 (0.06)	0.04 (0.00)	−0.72 (0.06)
963	−0.38 (0.07)	−0.06 (0.00)	−0.44 (0.07)	−0.45 (0.09)	−0.09 (0.00)	−0.55 (0.09)	−1.06 (0.09)	0.06 (0.00)	−1.00 (0.09)
964	−0.25 (0.02)	−0.09 (0.00)	−0.34 (0.02)	−0.11 (0.10)	−0.10 (0.00)	−0.21 (0.10)	−0.93 (0.02)	0.05 (0.00)	−0.88 (0.02)
971	−0.10 (0.02)	−0.11 (0.00)	−0.22 (0.02)	0.57 (0.21)	−0.15 (0.00)	0.43 (0.21)	−0.52 (0.07)	0.03 (0.00)	−0.49 (0.07)
972	−0.34 (0.07)	−0.07 (0.00)	−0.41 (0.07)	−0.08 (0.17)	−0.15 (0.00)	−0.23 (0.17)	−0.65 (0.08)	0.03 (0.00)	−0.62 (0.08)
973	−0.30 (0.02)	−0.11 (0.00)	−0.41 (0.02)	0.17 (0.13)	−0.12 (0.00)	0.05 (0.13)	0.11 (0.08)	0.03 (0.00)	0.14 (0.08)
974	−0.18 (0.02)	−0.10 (0.00)	−0.28 (0.02)	−0.11 (0.22)	−0.11 (0.00)	−0.22 (0.22)	−0.82 (0.08)	0.01 (0.00)	−0.81 (0.08)
981	−0.05 (0.02)	−0.11 (0.00)	−0.16 (0.02)	−0.24 (0.23)	−0.12 (0.00)	−0.35 (0.23)	−0.60 (0.12)	0.04 (0.00)	−0.56 (0.12)
Full sample	−0.20 (0.01)	−0.10 (0.00)	−0.30 (0.00)	−0.09 (0.05)	−0.11 (0.00)	−0.20 (0.05)	−0.70 (0.03)	0.04 (0.00)	−0.66 (0.03)

points. Despite the time-series variation, however, the cost of executing comparable orders using day crosses is significantly lower in every quarter. The results for after-hours crosses are similar, with the total cost differential over the entire sample at −20 basis points, but with substantially larger standard errors because of the small sample of after-hours crosses.

Execution cost differences between ECN-executed orders and broker-filled orders are larger. Over the entire sample period, the average implicit execution cost of ECN-executed orders is lower by 70 basis points, with a standard error of 3 basis points. Across quarters, the implicit cost differences are large and negative (ranging from −52 basis points to −106 basis points), with one exception. That exception is the third quarter of 1997, when the implicit cost differential is positive (11 basis points), but statistically insignificant (the standard error is 8 basis points). Since explicit cost differentials are marginally positive (4 basis points), the average total cost differential across all quarters is −66 basis points.

4.3.1. Matching issues and robustness

The matched sample methodology employed above has a number of obvious advantages. It does not presume linearity or any particular functional form in the relation between order characteristics and execution costs, and provides intuitive estimates of cost differences. However, the estimates are only as good as the matching criteria and precision of the matches. For example, it is possible that the execution cost differentials that we document above are simply due to differences in average price levels between orders executed in the four categories. Since execution costs are computed as percentages (of P_d), a large percentage of low-priced stocks in a category could result in substantially higher execution costs—entirely unrelated to the mechanism's ability to provide low-cost execution. To determine if this is the case, we compute the difference between the price of each order executed on an alternative trading system and the price of matching orders. The average price (percentage price) differences for crossed and ECN-executed orders (from matched broker-filled orders) are \$1.22 (4%) and \$0.30 (2%), respectively. These differences seem unlikely to be large enough to account for observed differences in execution costs.

The matching criteria employed above do not require broker-filled orders to be executed in the same quarter as the primary order. This provides considerable flexibility and allows for multiple matches. In fact, the median number of broker-filled orders that match day-crossed and ECN-executed orders are 166 and 57, respectively. However, given the quarter-to-quarter variation in sample sizes and order characteristics, it is possible that quarter-specific idiosyncratic effects account for observed execution cost differences. To determine if this is the case, we re-match orders with an additional criterion: we require that broker-filled orders execute in the same quarter. This reduces the median number of matches to 17 for day-crossed orders and eight for ECN-executed orders. For the full sample, the total execution cost difference between day-crossed and broker-filled orders changes only marginally, from –30 basis points to –31 basis points, and for ECN-executed orders, the differential changes from –66 basis points to –59 basis points.

Finally, we implement an extremely tight matching algorithm in an attempt to control for all stock-specific differences in trading. Specifically, we match each day-crossed or ECN-executed order with broker-filled orders in the *same* stock and in the *same* week. Using such restrictive matching conditions, we are only able to match 50% (30%) of the crossed (ECN-executed) orders and the median number of matches is one. Moreover, these matches occur most frequently for the very largest and most liquid stocks. For day crosses, the average market capitalization of this matched sample is \$20 billion, which is in the 97th percentile of the distribution of market values for all day crosses. Similarly, the average market capitalization of the matched sample for ECNs is \$10 billion, which falls in the 99th percentile of the distribution of market values for all ECN orders. Thus, the orders that are being matched represent a small, selective sample of very liquid stocks. While the average cost differential falls, it is still negative, at –9 basis points for crosses and –6 basis points for ECN-executed orders.

4.4. Regression-based evidence

An alternative method for estimating execution cost differentials is to estimate cross-sectional regressions of total execution costs on various control variables and indicator variables for orders executed on alternative trading systems. Following [Chan and Lakonishok \(1995\)](#) and [Keim and Madhavan \(1997\)](#), we use independent variables that capture well-known determinants of execution costs. Relative volume controls for the fact that orders that are large relative to normal trading volume are likely to have higher execution costs because of adverse selection effects. The inverse of the stock price accounts for price discreteness, and the logarithm of market capitalization takes into consideration the fact that liquidity is typically greater for larger firms. An exchange indicator (equal to one for NYSE/Amex firms and zero otherwise) is included because of evidence that transaction costs are typically higher on Nasdaq. The volatility of returns (measured as the standard deviation of daily returns over a ten-day period prior to the decision date) and the cumulative size-decile adjusted return (measured over the same interval) are also included in the regression to reflect market conditions. In particular, traditional trading mechanisms may be the liquidity provider of last resort, especially in fast-moving markets.

A limit order indicator variable equal to one for limit orders and zero for market orders is also included in the regressions, and we incorporate an interactive variable that allows for different average execution costs for ECN limit orders. Consequently, the regression estimates will control for the disparity in the use of limit orders across brokers and ECNs observed in [Table 2](#). Finally, in some specifications, we include indicator variables for each institution in the sample. [Keim and Madhavan \(1997\)](#) show that such fixed effects can be important because institutions have different approaches to filling orders that are reflected in their execution costs. These regressions are conservative in the sense that if an institution uses proportionately more of one trading platform, the regression ascribes the difference in average execution costs to the institution rather than the trading platform. We see a great deal of institution-specific variation in the use of trading platforms. ECN use tends to be heavier for some institutions; for example, 24 out of 59 institutions use ECNs for at least 10% of their orders. Fewer institutions use crossing systems, but some use it frequently—31 institutions never use crosses at all, and 6 institutions use day crosses for at least 10% of their orders. Our primary variables of interest are indicator variables for day crosses, after-hours crosses, and ECN-executed orders (broker-filled orders are captured by the intercept).

[Table 4](#) presents separate regressions for buyer- and seller-initiated orders. The first regression presents base estimates while the second includes institution-specific indicator variables. Focusing on the base regressions, the coefficients on day crosses and ECN indicator variables are negative for both buy and sell orders. For buys (sells), day crosses provide execution costs that are 31 (32) basis points lower than broker-filled orders. Similarly, ECN-executed buys (sells) are cheaper by 28 (66) basis points. These estimates are similar in magnitude to those produced by the matched sample procedure. Not surprisingly, inclusion of institution-specific fixed

Table 4

Total execution cost regressions

This table presents estimates of cross-sectional regressions of total execution costs on the variables listed below. The independent variables include the logarithm of market value of equity, the logarithm of relative volume (the number of shares traded divided by the geometric mean of daily trading volume over the previous five days), the inverse of the stock price on the day prior to the trading decision, an exchange indicator variable equal to one if the stock is listed on the NYSE/Amex, a limit order indicator variable equal to one for limit orders and zero for market orders, the standard deviation of daily returns from ten days prior to the decision date, the cumulative size-adjusted return from ten days prior to the decision date, and indicator variables for whether the order was executed via an external cross or on an ECN. Since there is no indicator variable for broker-filled orders, it is captured by the regression intercept. The second column in each category (buy vs. sell) includes institution-specific indicator variables. The sample consists of all single mechanism orders. *p*-Values appear in parentheses.

	Buys		Sells	
Intercept	1.043 (0.000)	0.513 (0.000)	2.001 (0.000)	1.833 (0.000)
Log (market cap)	−0.032 (0.000)	−0.071 (0.000)	−0.097 (0.000)	−0.072 (0.000)
Log (relative volume)	0.054 (0.000)	0.071 (0.000)	0.057 (0.000)	0.065 (0.000)
Inverse price	7.845 (0.000)	8.562 (0.000)	4.382 (0.000)	5.063 (0.000)
Exchange indicator	−0.062 (0.001)	−0.021 (0.412)	−0.149 (0.000)	−0.047 (0.006)
Standard deviation	5.930 (0.000)	3.171 (0.000)	6.858 (0.000)	4.549 (0.000)
Cumulative return	0.423 (0.000)	−0.091 (0.237)	−0.698 (0.000)	−0.631 (0.000)
Day cross	−0.311 (0.000)	−0.172 (0.000)	−0.325 (0.000)	−0.138 (0.000)
After-hours cross	−0.280 (0.001)	−0.186 (0.042)	−0.182 (0.023)	−0.006 (0.937)
ECN	−0.281 (0.000)	−0.280 (0.000)	−0.660 (0.000)	−0.539 (0.000)
Limit order indicator	−0.293 (0.000)	−0.147 (0.000)	−0.118 (0.000)	−0.639 (0.000)
Limit order * ECN	−0.294 (0.000)	−0.188 (0.000)	−0.081 (0.407)	−0.011 (0.911)
Institution indicators (fixed effects)	—	Yes	—	Yes
<i>N</i>	381,838	381,838	319,403	319,403
Adj- <i>R</i> ²	0.02	0.03	0.02	0.03

effects significantly lowers execution cost differentials. For buys (sells), the difference between day crosses and broker-filled orders falls to 17 (14) basis points. Similarly, for ECN-executed buy (sell) orders, the cost differentials fall to 28 (54) basis points. The coefficients on after-hours crosses are negative, although the differential for sell orders is not significant after the inclusion of fixed effects.

4.5. Endogeneity issues

The matched sample and regression-based approaches might adequately capture differences in order characteristics and difficulty, but could still miss an important economic component of the differences between the trading mechanisms. The more aggressive the trading mechanism, the more likely the order is to be filled. Thus, aggressive trading mechanisms are more likely to attract difficult-to-fill orders. Of course, the more difficult the order, the higher are the ex post execution costs. Since the tradeoffs are clear ex ante, these arguments suggest endogeneity in ex post execution costs. For example, observed execution costs for ECNs are conditional on the trader having chosen to route the trade to the ECN (based on expected costs). This implies that regressions of realized execution costs on order characteristics provide inconsistent estimates of the effect of alternative trading systems.

Madhavan and Cheng (1997) face a similar problem in analyzing executions in the upstairs versus downstairs market of the NYSE because traders endogenously choose between executing trades in the upstairs or downstairs market. They adapt and implement a simple two-stage econometric procedure that explicitly corrects for this self-selectivity. Since the econometrics of this procedure, known as endogenous switching regressions, are described in Madalla (1983, p. 283) and Madhavan and Cheng (1997, pp. 194–195), we do not reproduce it in detail here. However, we provide some basic intuition for the model to help motivate our specifications. The procedure consists of estimation of the following two-equation system:

$$y_i = \beta X_i + \gamma Z_i + \varepsilon_i, \quad (3)$$

$$Z_i^* = \theta W_i + u_i, \quad (4)$$

where

$$Z_i = \begin{cases} 1 & \text{if } Z_i^* > 0, \\ 0 & \text{otherwise,} \end{cases}$$

where y_i is the (total) execution cost of each order, X_i is a vector of variables controlling for order difficulty and characteristics (and includes a constant), W_i represents a vector of instruments, and Z_i is a binary variable equal to one if the order is executed on an alternative trading system (external crossing or ECN) and zero if the order is broker-filled. The procedure requires a first-stage structural probit regression that predicts the choice of trading mechanism as in Eq. (4). The second-stage regression, Eq. (3), then uses the estimates from the first-stage regression to provide consistent estimates of β and γ .

To implement the procedure, we create two subsamples: one with day crosses and broker-filled orders, and another with ECN-executed orders and broker-filled orders. The regressions are then estimated separately for these subsamples.⁵ Like most

⁵The small sample of after-hours crosses prevents us from estimating switching regressions for that subsample.

solutions to endogeneity problems, the first stage regression requires good exogenous instruments (W_i) for the system to be identified and well specified. Our choice of instruments is motivated by tradeoffs in the choice of trading venue and by estimation considerations. To that end, we identify factors important to the choice of trading venue so that the first-stage probit regressions produce a good fit. In the reported regressions, we use as instruments in the probit model an exchange indicator variable, three indicator variables for the investment style of the originating institution (momentum, diversified, and value), and return volatility. The exchange indicator variable picks up clientele effects of the sort described earlier, and is an important determinant of the trading mechanism used to execute an order. The investment style indicators account for the natural tendency of institutions with aggressive investment styles to use aggressive trading mechanisms. For example, an institution following a “momentum” trading style might be less likely to use passive crossing systems because of a potentially lower likelihood of the order being filled before its information expires. On the other hand, an institution with a value-based investment style, anticipating lower execution costs and being more patient, may be more likely to use passive crossing systems. Since we are missing style information for 14 institutions, the regression does not suffer from perfect collinearity. Finally, the volatility of return proxies for the potential benefit of considering different trading mechanisms. We perform a battery of robustness checks with different instruments in the first-stage regressions and alternative specifications of the second-stage regressions. For instance, using data from Instinet Inc., we use the proportion of total trading volume accounted for by Instinet as an instrument. In the second-stage regressions, we also include variables such as the cumulative return. These, and other such specifications, produce similar results and are not presented in tables.

Panel A of Table 5 shows the results of the first-stage regressions for the two subsamples. Since the magnitude of the coefficients is not easily interpretable, we also report marginal effects (changes in probabilities) in square brackets. For indicator variables (exchange listing and investment style indicators), the change in implied probabilities is computed assuming a change in the independent variable from zero to one. For the continuous variable (the volatility of return), the change in probability is computed assuming a one standard deviation change from the mean of the variable.

The first-stage regressions appear to be well specified. The pseudo- R^2 s range from 0.13 to 0.23 and all of the variables have statistically significant coefficients. More important, some of the implied probabilities are quite large. For buys, for instance, an order in a listed stock is 11% more likely to be crossed and 23% less likely to be executed on an ECN. Similarly, a one standard deviation increase in the volatility of return results in a large increase in the probability of choosing an ECN and a large decrease in the probability of choosing a crossing network.

Our primary interest is in the second-stage regressions shown in Panels B and C. As in Table 5, we use market capitalization, relative volume, and the price inverse as independent variables, and in Panel C, we include institution-specific indicator variables as well. The importance of the selectivity correction is given by the

Table 5

Endogenous switching regressions

Panel A shows the coefficients, marginal effects (in square brackets), and p -values of the first-stage regressions predicting the use of a day cross or an ECN. For indicator variables, the marginal effects (implied changes in probability) are computed assuming a change from zero to one. For continuous variables, the marginal effects are computed assuming a one standard deviation change from the mean. Panels B and C show the second-stage regressions with and without institution-specific indicator variables (fixed effects) that use the first-stage regressions to correct for endogeneity.

	Day crosses		ECN	
	Buys	Sells	Buys	Sells
<i>Panel A: Stage I (Probit) regressions</i>				
Intercept	−0.365 (0.000)	−0.428 (0.000)	−1.018 (0.000)	−1.133 (0.000)
Exchange indicator	0.768 [0.11] (0.000)	0.344 [0.04] (0.000)	−1.230 [−0.23] (0.000)	−0.956 [−0.12] (0.000)
Standard deviation	−2.729 [−0.51] (0.000)	−3.341 [−0.46] (0.000)	1.670 [0.23] (0.000)	3.517 [0.30] (0.000)
Momentum style indicator	−3.762 [−0.22] (0.000)	−3.063 [0.16] (0.000)	0.052 [0.02] (0.000)	−0.018 [−0.02] (0.000)
Value style indicator	−1.719 [0.16] (0.000)	−1.302 [0.09] (0.000)	0.411 [0.07] (0.000)	0.248 [0.03] (0.000)
Diversified style indicator	−1.271 [−0.26] (0.000)	−0.992 [0.15] (0.000)	0.495 [0.07] (0.000)	0.020 [0.02] (0.000)
Pseudo- R^2	0.23	0.15	0.18	0.13
<i>Panel B: Stage II regressions</i>				
Intercept	1.289 (0.000)	2.173 (0.000)	1.393 (0.000)	2.448 (0.000)
Log (market cap)	−0.041 (0.000)	−0.0111 (0.000)	−0.051 (0.000)	−0.126 (0.000)
Log (relative volume)	0.054 (0.000)	0.056 (0.000)	0.057 (0.000)	0.065 (0.000)
Inverse price	8.488 (0.000)	6.292 (0.000)	8.358 (0.000)	5.122 (0.000)
Day cross	−0.553 (0.00)	−0.662 (0.000)	—	—
ECN	—	—	−0.621 (0.000)	−0.679 (0.000)
λ	0.0015 (0.000)	0.0020 (0.000)	0.0010 (0.001)	−0.0002 (0.685)
<i>Panel C: Stage II regressions with institution indicators (fixed effects)</i>				
Intercept	1.589 (0.000)	2.166 (0.000)	15.131 (0.000)	2.724 (0.000)
Log (market cap)	0.003 (0.432)	−0.077 (0.000)	0.007 (0.197)	−0.081 (0.000)
Log (relative volume)	0.075 (0.000)	0.065 (0.000)	0.081 (0.000)	0.076 (0.000)

Table 5 (continued)

	Day crosses		ECN	
	Buys	Sells	Buys	Sells
Inverse price	9.166 (0.000)	6.886 (0.000)	9.117 (0.000)	5.858 (0.000)
Day cross	−0.162 (0.022)	0.306 (0.052)	—	—
ECN	—	—	−0.261 (0.000)	−0.453 (0.000)
λ	0.0033 (0.422)	−0.0024 (0.005)	−0.0002 (0.499)	−0.0004 (0.425)

significance of the new independent variable, λ , reported in Panels B and C.⁶ The estimates in Panel B show that controlling for selectivity, execution costs at ECNs are 62 basis points lower than at traditional brokers for buyer-initiated orders, and 68 basis lower for seller-initiated orders. For day crosses, buy (sell) orders have cost differentials of 55 (66) basis points. Thus, accounting for selectivity in the distribution of ex post execution costs provides estimates of cost differences that are similar in sign, and larger in magnitude, than the matching and regression-based procedures employed earlier. The inclusion of institution-specific variables in the second-stage regressions (Panel C) weakens these results somewhat: the execution costs for ECN buys and sells (at 26 and 45 basis points, respectively) are slightly smaller than OLS results, although they are still economically and statistically significant. The results for crossed orders are mixed. Buy orders are cheaper (by 16 basis points), while crossed sell orders are associated with an *increase* in costs of 30 basis points. Given the large sample size, however, the statistical significance of this cost differential is not large.

Heckman et al. (1997) analyze the sources of selection bias in a problem in which some sample elements receive a “treatment” while others do not. The analogous treatment in our sample is the choice of a particular alternative trading system. They show that the selection bias associated with differences in the characteristics (or support) of the sample elements is an important component of total measurement bias. Consequently, we also estimate the endogenous switching regression on matched samples of the orders, where the matching is identical to that used to construct Table 4. Although we do not report these results in the table, the differentials in the execution costs of alternative trading systems, compared to broker-filled orders, continue to be negative. ECN buys (sells) have execution costs that are 71 (74) basis points lower, and for day crosses, buys (sells) have execution costs that are 54 (65) basis points lower. Thus, these differentials are comparable to those reported in Table 5. Overall, after controlling for selection bias, the

⁶The independent variable λ is a function of the estimation of the probit Eq. (4). Specifically, $\lambda = [-\phi(\theta w_i)/(1 - \Phi(\theta w_i))]$, where ϕ is the standard normal distribution function and Φ is the cumulative normal distribution function.

differentials in execution costs for ECNs and crossing systems continue to be negative and are generally similar in magnitude to those observed in the OLS results.

4.6. Multiple mechanism orders: an exploratory analysis

Thus far, our analysis has focused on orders in which all trades are executed using a single trading mechanism. In this section, we shift our attention to multiple mechanism orders. In multiple mechanism orders, order characteristics are held constant across the trades, and the investor chooses not only how to break up the order, but where to place the trades and in what sequence the trading venues will be used. As described earlier in Table 1, there are 73,070 such orders, representing approximately 10% of the entire sample of orders, but over 47% of the dollar trading volume in our sample. The average order size is \$2.0 million, significantly larger than single mechanism orders and on average, 4.07 trades are required to fill these orders. The size of these orders would presumably imply that minimizing execution costs is more difficult and more desirable; the use of multiple trading mechanisms suggest that institutions are actively ‘shopping’ for the best trading venue. If so, we should expect to see the smallest differentials in execution costs across trading systems—in effect, the individual components of these multiple mechanism orders are more likely to be the ‘marginal’ trades.⁷

We start by examining the use of the four trading mechanisms in filling these orders. We calculate the number of trades required to fill each order and then describe the frequency (percentage) with which each trading mechanism is used in filling that order. This calculation is performed separately for each set of n -trade orders. In other words, the percentages for each n -trade order add up to 100. To parsimoniously characterize these results, we show these percentages for the first trade in the order, the last trade in the order, and the average of the interior trades in the order (all trades between the first and last trade). To conserve space, we only show results for up to ten-trade orders, representing 95% of all orders. Orders with more than ten constituent trades reflect similar patterns.

The results of these calculations are shown in Table 6 and provide some indication of the strategic use of the different trading systems. ECNs and after-hours crosses appear as the first, intermediate (interior), or last trade in an order with equal frequency. There is a substantial decline in the use of day crosses as the last trade in an order, relative to the first or interior trades, and a corresponding increase in the use of broker-filled orders. For example, for four-trade orders, the use of day crosses drops from 5.3% for the first trade to 3.7% for the last trade. Not surprisingly, since the percentages must add up to 100, the use of broker-filled orders increases from 12.7% to 14.4%. In longer trade sequences (greater than six-trade orders), there is some increase in the use of brokers in the second-to-last trade in the sequence.

If there is any urgency or additional difficulty in completing an order, as there may be for the last trade in an order, a crossing system might not be used since it is passive. Urgency requires a search mechanism, and both ECNs as well as traditional

⁷ We are grateful to the referee for this interpretation.

Table 6

Multiple mechanism order trading intensity

This table shows the use of the four trading mechanisms to fill multiple mechanism orders. Each major row represents the number of trades required to fill an order (ranging from two to ten). For each trading mechanism, the table shows the percentage of cases in which the trading mechanism is used to fill the trade. These percentages are shown separately for the first and last trade in the order. Since the number of interior trades (those between the first and last trade) varies, the average of these interior trades is shown. For example, for four-trade orders, day crosses are used to fill the first (last) trade in the order in 5.3% (3.7%) of all cases. Day crosses are used to complete interior trades in 4.9% of the cases.

Number of trades in order	Trading mechanism	First trade in order	Average of interior trades	Last trade in order
2	Day cross	14.3	—	10.3
	After-hours cross	2.8	—	2.7
	ECN	12.3	—	12.0
	Broker-filled	20.4	—	25.0
3	Day cross	8.4	7.0	5.3
	After-hours cross	1.9	2.7	1.9
	ECN	7.9	8.3	7.3
	Broker-filled	15.0	15.3	18.8
4	Day cross	5.3	4.9	3.7
	After-hours cross	1.4	2.1	1.4
	ECN	5.5	5.4	5.5
	Broker-filled	12.7	12.7	14.4
5	Day cross	4.1	3.7	2.6
	After-hours cross	1.1	1.6	1.0
	ECN	4.2	4.0	4.2
	Broker-filled	10.4	10.7	12.0
6	Day cross	3.1	2.8	2.1
	After-hours cross	0.9	1.3	0.9
	ECN	3.2	3.2	3.3
	Broker-filled	9.5	9.3	10.4
7	Day cross	2.7	2.5	1.7
	After-hours cross	1.0	1.1	0.7
	ECN	2.5	2.6	2.7
	Broker-filled	8.2	8.1	9.3
8	Day cross	2.2	2.2	1.9
	After-hours cross	0.6	0.9	0.6
	ECN	2.1	2.2	2.2
	Broker-filled	7.5	7.2	8.5
9	Day cross	1.8	1.8	1.1
	After-hours cross	0.5	0.8	0.5
	ECN	2.0	1.9	2.0
	Broker-filled	6.8	6.7	7.6
10	Day cross	1.8	1.7	1.1
	After-hours cross	0.6	0.7	0.5
	ECN	1.7	1.6	1.7
	Broker-filled	5.9	6.0	6.8

brokers provide such a search mechanism. To the extent that the search for counterparties on an ECN is restricted to participating institutions, however, the risk of not filling the trade could be larger. This suggests that traditional brokers are used to fill trades when the opportunity cost of *not* filling a trade is large. To examine this possibility, for each order we collapse adjacent trades executed under the same trading mechanism (i.e., two adjacent trades filled by an ECN would be regarded as one) and then calculate the number of unique sequences represented in the data. For example, a two-trade order in which the first trade is filled by an ECN and the second by a broker would be represented by the symbols “EB.” Similarly, if the first trade is filled by a day cross and the second trade is filled by a broker, the order would be represented by the symbol “XB.”

There are approximately 1,100 unique sequences represented in the 73,070 multiple mechanism orders; however, 95% of all orders involve fewer than five switches between trading mechanisms. There is some evidence that brokers tend to be last in completing an order—approximately 60% of the sequences end with “B” (i.e., the last trade is filled by a broker). For this 60%, the evidence is consistent with the conjecture in [Hendershott and Mendelson \(2000\)](#) that the dealer market will become the “market of last resort.” It is important to note, however, that their model is a single-period model capturing a static tradeoff, whereas the orders in our sample have a dynamic component—it is possible that an institution “learns” from previously executed trades in the same order.

We wish to compare the execution costs of broker-filled and alternative trading system trades. Calculating execution costs at the trade level is complicated by the strategic decisions of investors, who are presumably concerned with minimizing the execution cost of the *order*. Moreover, given clustering in the sequences, with brokers more frequently trading the last piece of the order, trade-level execution costs need to control for the position of the trade in the sequence, as well as order and trade difficulty. The complexity of the decision process, as seen in the number of sequences in our data, inevitably makes these controls imperfect. With this caveat in mind, we proceed to an examination of trade-level execution costs for multiple mechanism orders.

4.6.1. Trade-level execution costs

We compute execution costs for each trade as in Eqs. (1) and (2), but set the numerator in both equations to equal the trade price and commissions of a single trade, rather than a value-weighted average of all prices and commissions constituting the order. Using this measure of trade-level costs, we estimate multivariate regressions similar to those in [Table 5](#). Since the dependent variable is measured at the trade level, we use a number of trade-level controls, in addition to the order-level control variables used earlier. For instance, relative trade size, defined as the number of shares in the trade divided by the number of shares in the order, proxies for the information content and difficulty of the trade. In addition, two trade-level control variables capture the influence on market prices of earlier trades from the order. Specifically, we use the trade sequence number (a number equal to the chronological number of the trade in that order), and the cumulative lagged

implicit cost (defined as the price movement from P_d to the execution price of the prior trade in that order). We also include the number of switches between trading mechanisms (as described earlier), and a dummy variable equal to one if the last trade in the order was executed by a broker. As before, the regressions are estimated separately for buys and sells, with and without institution-specific effects, and the results are reported in Table 7.

Our primary interest is in indicator variables for whether the trade was executed using a day cross, an after-hours cross, or an ECN. The coefficients for day crosses and ECN-executed trades are negative and statistically significant in all regressions. For day crosses with institution-specific effects, the coefficients are -0.11 for buys and -0.07 for sells. For ECN-executed trades, the corresponding cost differentials are -18 basis points and -13 basis points, respectively. The coefficients for after-hours crosses are statistically insignificant. The dummy variable, which indicates that the last trade of the order was executed by a broker, has a positive coefficient for buy orders, which is both economically and statistically significant at 14 basis points (and 17 basis points when fixed effects are included.). For sell orders, the coefficients are statistically insignificant. Thus, there is some evidence that for buy orders, brokers handle the last and most difficult part of the order. Even allowing for this, the cost differences that exist for single mechanism orders also appear, albeit to a smaller extent, at the trade level for multiple mechanism orders.

4.7. Exogenous shocks to alternative trading systems

Our sample period spans two major changes to trading in securities markets that are particularly relevant for our analysis in that they may provide exogenous shocks to the benefits of trading on alternative trading systems. The first change is Nasdaq's introduction of new order handling rules in 1997. These rules required that public limit orders be displayed in the best bid and offer (BBO) on Nasdaq; in addition, market makers were prevented from posting one quote on Nasdaq and a different quote on an ECN. The new rules effectively reduced fragmentation between the markets by integrating ECN prices into Nasdaq. By exposing ECN quotes to a wider market, these rules could have the effect of reducing the comparative advantage of trading on an ECN; see Barclay et al. (1999) for a complete discussion of the new rules. The second change in our sample period is the SEC-mandated change in tick size from eighths to sixteenths. Before the tick size change, investors were able to trade at finer price increments (thirty-seconds, sixty-fourths, or smaller) on ECNs but not on other trading venues. If there is an advantage to using finer price increments then this change should also reduce the comparative benefit of trading on an ECN; see Harris (1997) for a summary of the advantages and disadvantages of finer tick sizes.

4.7.1. Order handling rules

Nasdaq stocks were phased into the new order handling rules in 22 separate waves from January 20, 1997 to October 13, 1997. We obtained a dataset from Nasdaq that details the phase-in date for each individual stock and match this dataset with our

Table 7

Multiple mechanism order trade-level execution cost regressions

This table presents cross-sectional regressions of trade-level total execution costs (in percent) on order- and trade-level variables. Order-level variables include the logarithm of market value of equity, order size (the number of shares in the order divided by shares outstanding), the inverse of the stock price on the day prior to the trading decision, and an exchange indicator variable equal to one if the stock is listed on the NYSE/Amex (zero otherwise). Trade-level variables include relative trade size (number of shares in the trade divided by the number of shares in the order), trade sequence (the sequence number of trade in the order), the cumulative lagged implicit execution cost (defined as the price movement from the decision price P_d to the price of the prior trade in the order), and the percentage of the order already filled. Indicator variables for a day cross, after-hours cross, or an ECN trade are also included. p -Values appear in parentheses.

	Buys		Sells	
Intercept	2.921 (0.000)	3.327 (0.000)	2.656 (0.000)	5.659 (0.000)
Log (market cap)	−0.184 (0.000)	−0.109 (0.000)	−0.142 (0.000)	−0.082 (0.000)
Log (relative volume)	0.043 (0.000)	0.069 (0.000)	0.244 (0.000)	0.272 (0.000)
Inverse price	12.127 (0.000)	16.038 (0.000)	−1.299 (0.002)	1.264 (0.002)
Exchange indicator	−0.298 (0.000)	−0.051 (0.219)	−0.293 (0.000)	−0.001 (0.985)
Return volatility	12.767 (0.000)	1.170 (0.285)	21.715 (0.000)	16.153 (0.000)
Cumulative return	3.362 (0.000)	−0.051 (0.748)	−0.655 (0.000)	−0.276 (0.091)
Relative trade size	−0.454 (0.000)	−0.561 (0.000)	−0.217 (0.000)	0.302 (0.000)
Trade sequence	0.169 (0.000)	0.161 (0.000)	0.063 (0.000)	0.048 (0.000)
Cumulative lagged implicit cost	0.164 (0.000)	0.148 (0.000)	0.077 (0.000)	0.065 (0.000)
Percentage of order already filled	0.012 (0.859)	−0.141 (0.040)	−0.484 (0.000)	−0.367 (0.000)
Number of switches	−0.090 (0.000)	−0.035 (0.000)	−0.009 (0.211)	0.021 (0.006)
Last trade broker indicator	0.145 (0.000)	0.171 (0.000)	0.044 (0.205)	−0.095 (0.791)
Day cross	−0.079 (0.000)	−0.111 (0.010)	−0.057 (0.009)	−0.0671 (0.013)
After-hours cross	−0.107 (0.010)	−0.177 (0.008)	0.010 (0.415)	0.069 (0.287)
ECN	−0.215 (0.000)	−0.183 (0.000)	−0.235 (0.000)	−0.134 (0.000)
Institution indicators (fixed effects)	—	Yes	—	Yes
N	216,582	216,582	171,563	171,563
Adj- R^2	0.03	0.08	0.02	0.05

sample. To determine if the order handling rules influence the use of alternative trading systems, we restrict our sample to 45 institutions with at least 200 orders before and after the phase-in dates. This ensures robust pre-post comparisons. For each institution, we subtract the percentage of single mechanism orders executed by day crosses and ECNs before the phase-in dates from the percentage of orders executed using the same mechanism after the phase-in dates. The average difference is 1.7% for day crosses and -6.2% for ECNs, with corresponding paired *t*-statistics of 0.95 and 2.94, respectively. We also examine changes in various order characteristics before and after the order handling rules for this subsample but find no systematic differences between the pre- and post-period. Thus, the introduction of the new rules reduced the proportionate use of ECNs by institutions in our sample, but did not affect the use of crossing systems.

To examine whether the introduction of the new rules influences the cost differentials between alternative trading systems and traditional brokerage, we estimate regressions similar to those in Table 4. We eliminate all listed stocks from the sample because the order handling rules only pertain to Nasdaq stocks. We also remove any order in which the phase-in date for the stock falls between the decision date and the last trade of the order. This filter allows for “clean” pre-post comparisons. We then include three other indicator variables in the regressions. First, we construct an order handling rule indicator which is equal to one (zero) if the order was executed after (before) the phase-in date. This shows the effects of order handling rules on institutional trading costs in general. Second, we include post-period (pre-period) ECN indicators which are equal to one if an ECN order was executed after (before) the introduction of the new rules. These variables permit direct comparisons of ECN-executed orders before and after the order handling rules with the omitted category (broker-filled orders). As before, the regressions are estimated separately for buys and sells, with and without institution-specific effects.

Across all four regression specifications, the pre-period ECN indicator is large and negative, consistent with the results in Table 4. The post-period ECN indicator drops significantly in magnitude from the pre-period. For example, in the base regressions, the ECN cost differential falls from 90 basis points to 16 basis points for buys and from 117 basis points to 86 basis points for sells. Similar declines are evident in the regressions with institution fixed effects. This suggests that the new order handling rules caused a decline in the comparative advantage of trading on ECNs. However, the post-period cost differentials between ECNs and broker-executed orders remain negative, suggesting that the advantage is not completely eliminated.

4.7.2. Tick size changes

Unlike the order handling rules, tick size changes were implemented across all securities in a given exchange on a certain date for each exchange: the NYSE on June 24, 1997, Amex on May 7, 1997, and Nasdaq on June 2, 1997. We follow a procedure similar to that for order handling rules to determine if the use of alternative trading systems declined following the change in tick size. As before, we begin by restricting the sample to institutions with data both before and after the tick size change and compute the average change in the percentage of single mechanism orders executed

by day crosses and ECNs. The average difference is 2.2% for day crosses (paired t -statistic of 1.5) and -7.9% for ECNs (paired t -statistic of 12.4). As before, we also examine changes in various order characteristics before and after the tick size change but find no systematic differences between the pre- and post-period.

To determine if the tick size change influences cost differentials, we estimate regressions in Panel B of Table 8 similar to those for order handling rules. We do not use exchange listing as an exclusionary criteria but eliminate orders in which the date of the tick size change falls between the decision date and the last trade in the order. In addition to the control variables in Panel A of Table 8, we also include an

Table 8

The effects of order handling rules and tick size changes on cost differentials

This table presents estimates of cross-sectional regressions of total execution costs (in percent) surrounding changes in order handling rules (Panel A) and tick sizes (Panel B). The independent variables include those employed in Table 5 (logarithm of market value of equity, the logarithm of relative volume, the inverse of the stock price, an exchange indicator, the standard deviation of daily returns, and the cumulative size-adjusted return). Panel A regressors also include an order handling rule indicator equal to one (zero) if the order was executed after (before) the change in the order handling rules and a pre-period (post-period) ECN indicator equal to one if the order was executed prior to (after) the change in the order handling rules on an ECN. Panel B regressors include a tick size change indicator equal to one (zero) if the order was executed after (before) the change in the tick size, and pre-period (post-period) day cross, after-hours, and ECN indicators that are equal to one if the order was executed in the period before (after) the tick size change using a day cross, after-hours cross, or ECN respectively. Since there is no indicator variable for broker-filled orders, it is captured by the regression intercept. The sample for Panel A consists only of single mechanism orders in Nasdaq stocks. The sample for Panel B consists of all single mechanism orders. p -Values appear in parentheses.

	Buys		Sells	
<i>Panel A: Order handling rule regressions</i>				
Intercept	2.216 (0.001)	0.624 (0.016)	3.311 (0.000)	2.081 (0.000)
Log (market cap)	−0.122 (0.000)	−0.001 (0.981)	−0.015 (0.000)	−0.033 (0.070)
Log (relative volume)	0.074 (0.000)	0.108 (0.000)	0.190 (0.000)	0.209 (0.000)
Inverse price	6.449 (0.000)	8.429 (0.000)	6.100 (0.090)	2.442 (0.000)
Standard deviation	6.799 (0.000)	2.631 (0.007)	13.832 (0.000)	10.370 (0.000)
Cumulative return	0.624 (0.045)	−0.330 (0.053)	−0.958 (0.000)	−0.898 (0.000)
Order handling rule indicator	−0.164 (0.007)	−0.062 (0.366)	−0.206 (0.000)	−0.160 (0.022)
Pre-period ECN indicator	−0.901 (0.000)	−0.657 (0.000)	−1.171 (0.000)	0.723 (0.000)
Post-period ECN indicator	−0.164 (0.002)	0.141 (0.070)	−0.863 (0.000)	−0.524 (0.000)
Institution indicators (fixed effects)	—	Yes	—	Yes

Table 8 (continued)

	Buys		Sells	
<i>Panel B: Tick size change regressions</i>				
Intercept	1.211 (0.000)	0.564 (0.000)	2.226 (0.000)	1.775 (0.000)
Log (market cap)	−0.048 (0.000)	−0.001 (0.840)	−0.111 (0.000)	−0.073 (0.000)
Log (relative volume)	0.039 (0.000)	0.064 (0.000)	0.057 (0.000)	0.075 (0.000)
Inverse price	7.870 (0.000)	8.955 (0.000)	4.521 (0.000)	5.701 (0.000)
Standard deviation	6.499 (0.000)	3.242 (0.007)	6.443 (0.000)	4.002 (0.000)
Cumulative return	0.681 (0.000)	0.010 (0.915)	−0.813 (0.000)	−0.747 (0.000)
Exchange indicator	−0.081 (0.000)	−0.022 (0.269)	−0.139 (0.000)	−0.037 (0.054)
Tick size change indicator	0.029 (0.111)	0.214 (0.016)	−0.039 (0.160)	0.185 (0.072)
Pre-period ECN indicator	−0.801 (0.000)	−0.657 (0.000)	−0.888 (0.000)	−0.739 (0.000)
Post-period ECN indicator	−0.238 (0.000)	−0.112 (0.050)	−0.576 (0.000)	−0.247 (0.000)
Pre-period day cross indicator	−0.365 (0.000)	−0.177 (0.001)	−0.344 (0.000)	−0.232 (0.000)
Post-period day cross indicator	−0.268 (0.020)	−0.151 (0.070)	−0.530 (0.080)	−0.244 (0.060)
Pre-period after-hours cross indicator	−0.336 (0.002)	−0.210 (0.060)	−0.217 (0.042)	−0.075 (0.483)
Post-period after-hours cross indicator	0.075 (0.778)	−0.065 (0.811)	−0.540 (0.023)	−0.452 (0.060)
Institution indicators (fixed effects)	—	Yes	—	Yes

exchange indicator, a tick size change indicator equal to one (zero) if the order was executed after (before) the tick size change date, and separate pre- and post-indicator variables for day crosses, after-hours crosses, and ECNs.

The tick size change indicator is positive across all specifications and significant in one out of four regressions. Therefore, consistent with Jones and Lipson (2001), we find no evidence that the move to sixteenths reduced trading costs for institutions and some evidence that costs may actually have increased. However, our focus is primarily on cost differentials between alternative trading systems and broker-executed orders. Here, the results mirror those for the order handling rules. There appears to be little or no change in cost differentials for day or after-hours crosses. For ECNs, all four regression specifications show a decline in the cost advantage to ECNs after the tick size change. However, the cost advantage does not completely disappear, as the post-period ECN indicator remains negative.

Overall, our results indicate a reduction in the estimated cost advantage to ECNs over time. This reduction is consistent with a competitive market for order flow, combined with regulatory changes that are designed to minimize the fragmentation of markets. It is possible, as discussed in the introduction, that the differentials we find represent a characteristic of order difficulty for which we have not controlled, or a benefit to broker trading that we have not considered. Alternatively, we speculate that the remaining (economically significant) differential will continue to diminish over time.

5. Conclusion

This paper examines the execution costs of trades sent to traditional and alternative trading systems. In a sample of \$1.6 trillion in trades executed by 59 different institutions between the first quarter of 1996 and the first quarter of 1998, we find that there is substantial variation in the trading venues used by these institutions as well as in the execution costs of their trades. We control for differences in order characteristics using both a matching technique and a regression analysis and find that orders sent to traditional brokers have higher execution costs than those executed by alternative trading systems. Crossing systems also have lower realized execution costs than broker-filled orders. However, this trading platform has the lowest fill rate in our sample and the estimated fill rates for single-mechanism orders are likely upward biased. Consequently, while we are able to control for some aspects of the unfilled portions of orders, it is important to emphasize that the execution cost differences we report could still be conditional on execution. ECNs, which aid in price discovery, have the lowest execution costs.

We also control for endogeneity in the trading venue selection using an endogenous switching regression, but the rank ordering of execution costs across trading platforms persists. Further, in a sample of multiple mechanism orders, in which institutional investors make strategic use of different trading platforms to minimize execution costs, we continue to find that trading costs are lower at alternative trading systems. In all cases, the inclusion of institution-specific fixed effects reduces these differences, but with the exception of one case, it does not eliminate them.

In equilibrium, average trading costs across different trading venues should be equal. There are at least two possible explanations for the cost differences that we observe. The differentials could simply be due to a cost of trading that we have not accounted for, or reflect a benefit of broker trading that we do not measure. Indeed, in the case of crossing systems, the opportunity cost of unfilled orders could explain the differences. Another possibility is that the differentials are temporary, as markets move from one equilibrium to another. This explanation could be particularly pertinent for ECNs because the nature of their competition with Nasdaq market makers has changed over time. We examine two specific regulatory shocks to trading on ECNs over our sample period that could reduce the advantage of trading on ECNs—the 1997 order handling rules and the SEC-mandated reduction in tick sizes.

We find a reduction in ECN use and cost differentials subsequent to both these events.

Since the end of our sample period, price increments have been further reduced to pennies as part of the decimalization of the financial markets. One might anticipate changes in institutional order-routing patterns subsequent to decimalization, similar to those following the 1997 tick size changes. However, the impact of decimalization is considerably more complex. It is cheaper for traders to “step ahead” of institutional orders when the price increment is a penny than when it is a tick (widely known as being “pennied”). Harris (1999) argues that to avoid such front running, institutions might increase their use of alternative trading systems. On the other hand, if institutions benefit from the availability of finer price increments on ECNs, then decimalization eliminates that advantage and would reduce order flow to ECNs. The net effect is ultimately an empirical issue and two recent studies provide preliminary evidence on the subject. An internal study by the Plexus Group (see Wagner, 2001) finds no evidence of an increase in institutional trading costs as a result of decimalization. A Nasdaq (2001) report finds a decline in ECN usage subsequent to decimalization and concludes “buy-side traders may route relatively more orders to institutional brokers than ECNs.” Since sub-penny trading is also allowed in some stocks (making it even less costly to step ahead of an institutional order), some ECNs such as Bloomberg Tradebook have banned participants from entering orders with increments less than one cent. The changes described above reflect an evolution of the marketplace where, as competitors react and as regulatory policies are changed, the cost differentials across trading platforms have diminished.

Appendix

The following algorithm categorizes all trades into one of four categories.

1. A *day cross* is identified if either of the following two conditions are satisfied:
 - (a) The broker ID is marked “POSIT,” thus identifying the use of the POSIT crossing mechanism. The commissions for trades on POSIT are two cents per share.
 - (b) The broker ID is marked “ITG” and the commissions are two cents per share. Early in the sample period, some institutions use ITG as the broker ID even though the trades are crossed on POSIT; this filter isolates such trades and categorizes them as a cross.
2. An *after-hours cross* is identified if either of the following two conditions are satisfied:
 - (a) The broker ID is marked “CROSS,” thus identifying the use of (after-hours) Instinet Crossing. The commissions for trades using Instinet’s crossing operation are one cent a share.
 - (b) The broker ID is marked “Instinet” and commissions are exactly one cent per share.

3. All trades with a broker ID marked “Instinet” without commissions of one cent per share are regarded as *ECN-executed* trades.
4. Trades not filled by the above three mechanisms are regarded as *broker-filled*.

Regulation ATS, adopted by the SEC in 1998, requires that alternate trading systems either register as an exchange or remain a broker–dealer. Since the majority of such organizations have chosen not to register as exchanges, they appear in our data as brokers. As a result, the classification procedure described above relies on broker identification. The classification algorithm also exploits regularities in commission schedules. For example, in 1(b) above, when the broker ID is marked “ITG” we rely on commissions to isolate crossing (POSIT) trades and attribute the remainder to ITG’s full service brokerage operations. We do not believe this results in a serious bias. Commissions on the trades marked “POSIT” are *always* two cents per share, suggesting that using commissions to classify trades marked “ITG” is appropriate. Only 10% of all trades marked “ITG” or “POSIT” are attributed to full service brokerage. In the case of Instinet, commissions of one cent per share are similarly used to isolate crossing trades not directly marked as such (see 2(b) above). This procedure influences only 5% of all trades in which Instinet (either “Instinet” or “Cross”) is the marked broker ID. Even given the above qualifications, it is possible that there is some mixing of trades across categories. For example, it is possible that some traditional brokerage trades are classified as ECN trades. Conversations with the Plexus Group indicate that the magnitude of such potential misclassification is quite small. Even if there is some degree of misclassification, such error would reduce (rather than increase) systematic variation in order characteristics and costs across the four categories.

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